# Genomic Mechanisms that Facilitate Proliferation of Multiple Drug Resistant "superbugs" : a case study using Serratia as a model organism

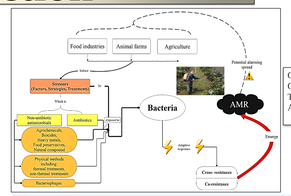
Adi Chauhan, Aryan Taywade and Abhinav Balaji

## Abstract

Exposure of environmental bacteria to long-term pollution can facilitate recruitment of multidrug-resistant (MDR) genes; these bacteria are referred to as "superbugs", leading to potential pathogenic outbreaks. To assess the development of superbugs in the environment, we worked on a bacterial genus- Serratia species: which are either pathogenic or beneficial to the environment. A central question in our research was: can whole genome sequencing be used as tool to obtain a comprehensive understanding of MDR in pathogenic bacteria? This approach can facilitate molecular diagnostics on the presence of pathogenic bacteria, enable novel drug design and prevent pathogenic outbreaks like the COVID-19 pandemic. For this project, we advanced our research using a recently isolated Serratia marcescens species strain SRS-8-S-2018 as a model organism, shown to resistant multiple antibiotics. Further, we used several computational methods to understand the traits of our model organism. A genomic map of strain SRS-8-S-2018 revealed several specialty genes for antibiotic resistance, virulence factors, drug targets, and transporters- all of which can render MDR capabilities to this strain. Notably, all the 5 classes of efflux pumps were identified in this model organism, which are: ATP-binding cassette (ABC) superfamily, (MFS), (MATE) family, the small multidrug resistance (SMR), (RND) superfamily. Moreover, an arsenal of β-lactam antibiotic resistance system was also identified in the strain- all these can be good targets to block drug efflux in opportunistic pathogens. We then developed a better understanding of the protein structure of a representative of ABC transporter, which are dominant in this strain; the 3-D protein model demonstrated conserved domains and protein families which can be the basis for effective newly designed antipeptide drugs. Overall, this study advances our understanding on MDR in a model opportunistic pathogen, thus paving the way for precise drug design to mitigate bacterial antibacterial resistance.

## Introduction

- Multi-drug resistance (MDR) genes are being acquired by environmental bacteria- known as **"superbugs"** and has become a rampant worldwide problem. **These MDR pathogens cause bacterial infections and sepsis, which is the leading cause of deaths in hospitals.** Thus, controlling the emergence and spread of MDR warrants immediate attention and further research.
- This emergent issue has been brought to the forefront by both scientists and popular media; as shown in the figure emphasizing that along with the public health measures taken for the COVID-19 pandemic, we should not lose sight of the global MDR issue.
- As a continuation of our science fair project from last year, we investigated a group of bacteria called *Serratia* species; these have evolved to be either pathogens (medical-type) or the environment types which are beneficial. **However, our previous project revealed that our model environmental *Serratia* strain SRS-8-S-2018 was functionally closer to a medical pathogenic type, despite being isolated from the environment, suggesting the MDR genes are being spread into the environment and superbugs are evolving.**
- This finding is supported by the fact that environmental pollutants drives genomic reshuffling and evolution of MDR in microorganisms such as *Serratia*. However, information is lacking on the genomic mechanisms by which these opportunistic pathogens are evolving as superbugs, thus posing serious public health concerns.

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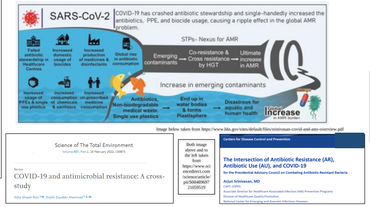



Image below taken from https://www.bbc.gov/news/default/files/strategic-covid-and-amr-overview.pdf

Science of The Total Environment
volume 807, Part 3, 10 February 2022, 150873
Review
COVID-19 and antimicrobial resistance: A cross-study
Sofia Graça Seiça ⚑, Vítor Zapata Abarca ⚑ & W

Both image above and to the right taken from https://www.sciencedirect.com/science/article/pii/S0048969721058851

Centers for Disease Control and Prevention
The Intersection of Antibiotic Resistance (AR), Antibiotic Use (AU), and COVID-19
For the Presidential Advisory Council on Combating Antibiotic-Resistant Bacteria
Arjun Srinivasan, MD
April 14th
Associate Director for Healthcare Associated Infection (HAI) Prevention Programs
Division of Healthcare Quality Promotion
National Center for Emerging and Zoonotic Infectious Diseases

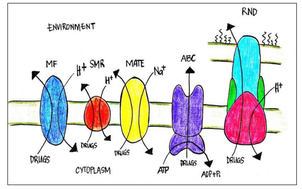



Figure on the left is taken from: https://microbewiki.kenyon.edu/index.php/Serratia_marcescens%27_different_drug_resistant_efflux_pumps

- **One mechanism used by bacteria to fight against antibiotics is via the presence of efflux pumps,** which enables the cells to self regulate their internal environment by expelling out antimicrobial agents.
- Five families of bacterial efflux pumps are currently known - the ATP-binding cassette (ABC) superfamily, the major facilitator superfamily (MFS), the multidrug and toxic compound extrusion (MATE) family, the small multidrug resistance (SMR) family, and the resistance-nodulation-division (RND) superfamily.
- **Further, β-lactam antibiotics (such as penicillin, cephalosporins and cephamycins, monobactams, carbapenems and carbacephems), are 65% of the total antibiotics market and the most prescribed antibiotic (https://www.ncbi.nlm.nih.gov/books/NBK545311/), thus driving MDR.**
- However, molecular mechanisms for MDR and beta-lactam resistance in opportunistic pathogens are not fully known.

- Therefore, In this project, we utilized in-silico computational tools to obtain a deeper understanding on the genomic mechanisms of drug resistance in *Serratia* species, which serves as a good model due to their opportunistic pathogenic traits.
- **Our project is noteworthy because the world is currently facing a severe pandemic caused by the COVID-19 virus.** The rampant emergence of superbugs also have the potential to cause global pandemics but have not received as much attention.
- This study can pave the way to identify or design targets that can effectively block efflux pump genes of pathogenic bacteria, thus preventing their spread in the environment and prevent hospital infections.

## Overarching Research Questions

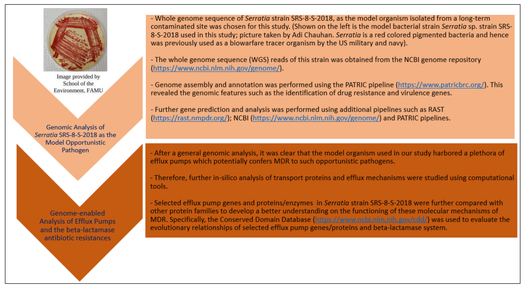
- What are the molecular mechanisms that exist in opportunistic bacterial pathogens that render multiple drug resistance?
- Can genomic workflows facilitate a better understanding of the molecular basis of MDR?
- Which MDR traits will likely be ideal targets for designing novel antibiotics to combat the global spread of superbugs.

## Objectives

**Our main objectives are as follows:**

- Use *Serratia* spp. as a model organism to test a variety of computation tools and identify the molecular mechanisms of MDR;
- Because efflux pumps and beta-lactamase mechanisms render MDR traits, what is the genomic basis of these traits in the model organism?
- Can genomic methods serve as a precise tool to design better drug targets to block the extrusion of antibiotics?

## Materials and Methods




## Results and Discussion

**Experiment #1:** Obtain a genomic map of strain SRS-8-S-2018 by using the the Pathosystems Resource Integration Center (PATRIC pipeline; https://www.patricbrc.org/.

This mapping was done to visualize the presence of genomic features in our model bacterial strain

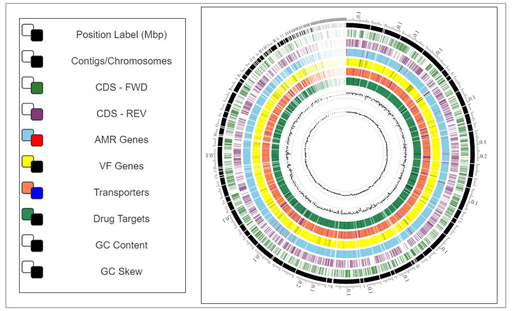


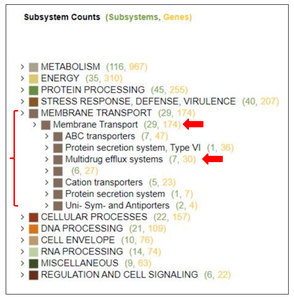

**Fig. 1.** Shown is a circular genome map of strain SRS-8-S-2018, obtained using the default settings within the PATRIC pipeline. The map consists of 8 circles from inside out; first shows size in Mbp, followed by DNA contigs, AMR Genes, VF Genes, Transporters, Drug Targets, GC Content, and the GC Skew.

**This revealed specialty genes for antibiotic resistance (63), virulence factors (93), drug targets (165), and transporters (261)- all of which can render MDR capabilities to this strain.** This also indicated that this environmental type strain of *Serratia* has acquired multiple virulence and antibiotic traits, which warrants further studies.

**Fig. 1.** Produced by Adi Chauhan, Aryan Taywade and Abhinav Balaji using PATRIC (https://www.patricbrc.org/).

## Results and Discussion

**Experiment #2:** Computational analysis using Rapid Annotation using Subsystems Technology (RAST) and PATRIC pipelines to further examine the efflux pumps/ membrane transport proteins as mechanisms for MDR traits.



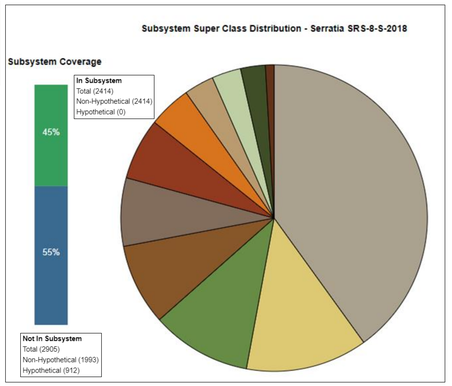




**Fig. 2.** Metabolic model obtained using the **RAST in-silico workflow revealed that only 45% of the genes could be assigned to known functions with 55% with no affiliation to known functions. This indicates that genomic technologies still cannot fully predict the molecular composition of an opportunistic pathogen such as the one used in our study.** It is likely that contaminated soil environments are driving evolution of opportunistic bacteria in the environment with genomic traits that remain complex for biological interpretation.

- **Membrane transport was identified as one major gene class in this strain, which renders MDR traits via their efflux mechanisms.**
- Interestingly, **all the 5 known classes of efflux pumps were identified in this model organism**, which are: ATP-binding cassette (ABC) superfamily, the major facilitator superfamily (MFS), the multidrug and toxic compound extrusion (MATE) family, the small multidrug resistance (SMR) family (a subgroup of the drug/metabolite transporter superfamily), and the resistance-nodulation-division (RND) superfamily.
- **Hence this took us to the next question- what is the molecular structure of these membrane transport systems, specifically the most dominant ABC type**. A complete understanding of the MDR traits can lead to the development of better drug targets, and our study is filling this knowledge gap.

**Figure 2.** Produced by Adi Chauhan, Aryan Taywade and Abhinav Balaji using RAST=Rapid Annotation using Subsystems Technology (https://rast.nmpdr.org/seedviewer.cgi?page=Home).

**Experiment #3:** Identify the genomic structure of ABC transporters in our model organism relative to other *Serratia* spp.

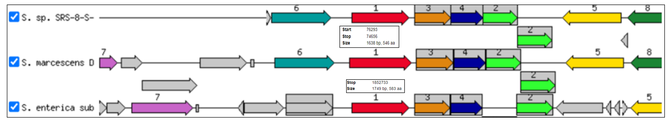



**Figure 3.** We compared the genomic region of a dipeptide-binding ABC transporter in our model organism compared with two other *Serratia* species using RAST=Rapid Annotation using Subsystems Technology (https://rast.nmpdr.org/seedviewer.cgi?page=Home). The focus gene is red and numbered 1; genes whose relative position is conserved in two other species are functionally coupled and share gray background boxes.

**Our strain ABC gene was of 1638 bp in size compared to 1749 bp in *S. enterica*.** This ABC gene system was then evaluated using 3D modeling to further understand the molecular structure.

**Fig 3.** Produced by Adi Chauhan, Aryan Taywade and Abhinav Balaji using RAST=Rapid Annotation using Subsystems Technology (https://rast.nmpdr.org/seedviewer.cgi?page=Home).

**Experiment #4:** Assess the protein 3-D structure of an ABC transport protein family, which is the largest membrane transport system in our model organism.

Domains are shown in different colors and amino acid residues are shown in yellow. The arrows on the strands point in the N-to-C N-terminus, because the protein is translated from messenger RNA and created from N-terminus to C-terminus.

**Such detailed molecular analysis can reveal protein functional sites that can be blocked by newly designed antibiotic drugs**, such as antimicrobial peptides, to combat the superbugs in environmental and hospitals.

**Fig. 4.** Produced by Adi Chauhan, Aryan Taywade and Abhinav Balaji using NCBI (https://www.ncbi.nlm.nih.gov/Structure/icn3d/full.html?&mmdbid=107503&bu=1&showanno=1&source=full-feature)

**Experiment #5:** Computational genomic analysis of our model organism for the presence of β-lactam resistance.

- Bacterial resistance to β-lactam antibiotics is growing and is a global public health problem.
- β-Lactam resistance is mainly due to 1) **enzyme β-lactamases,** which breaks the lactam ring or 2) or by **modification of their targets in the cell wall (penicillin-binding Proteins, PBPs)**; it is therefore desirable to characterize in our model organism.

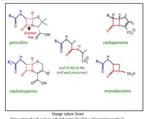



Image taken from https://microbewiki.kenyon.edu/index.php/antimicrobials

**Fig. 5.** Our computational analysis found **a predominance of β-lactamase enzymes, drug efflux pumps, penicillin binding proteins and reduction of porins- all these features in conjunction is providing MDR** to our model organism and likely serve as good targets for new drugs that can block these mechanisms.

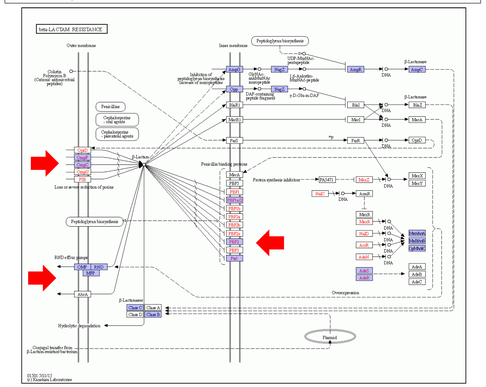



**Fig. 5.** Produced by Adi Chauhan, Aryan Taywade and Abhinav Balaji using RAST=Rapid Annotation using Subsystems Technology (https://rast.nmpdr.org/seedviewer.cgi?page=Home).

## Conclusions

- Multi-drug resistance (MDR) traits are being acquired by environmental bacteria- known as **"superbugs"**, which has become a worldwide public health problem. Overall, this study advances our understanding on MDR in a model opportunistic pathogen, thus paving the way to design novel antipeptide drugs to mitigate bacterial infections.
- This study was focused on an emerging "superbug" with MDR- the *Serratia* species that causes bloodstream infections, pneumonia, urinary tract infections, central nervous system infections, and conjunctivitis. Therefore, **obtaining a deeper understanding on the genomic mechanisms of MDR is critical** such that better drug targets can be designed to mitigate this global health issue.
- To this end, several computational methods were utilized to understand the metabolic traits of a model pathogenic bacteria belonging to the *Serratia* genus (strain SRS-8-S-2018). Genomic map of strain SRS-8-S-2018 revealed the presence of multiple genes (numbers in parenthesis), for antibiotic resistance (63), virulence factors (93), drug targets (165), and transporters (261)- all of which can render MDR capabilities to bacterial pathogens.
- Furthermore, **computational analysis revealed the presence of all 5 known classes of drug efflux pumps in our model organism,** providing clues on its versatile MDR mechanisms.
- A better understanding of the protein structure of a representative of the dominant ABC transporter was obtained by generating a 3-D model. This identified domains and amino acid composition of the ABC transporter in our model organism; **such analysis reveals protein sites for effective drug targeting for future research.**
- **Mechanism for β-lactam resistance was better understood**; this is relevant because β-lactam antibiotics are the most prescribed antibiotic causing rise of superbugs.
- The next step is to **identify mitigation strategies** to combat the MDR problem. For example, we can engage our peers to collaborate on efforts related to **antibiotic- control and surveillance**, such as being involved with a nationwide project called **PARE- Prevalence of Antibiotic Resistance in the Environment** (PARE; https://sites.tufts.edu/ctse/pare/) or the **Emory antibiotics resistance center** (https://antibiotics.emory.edu/).